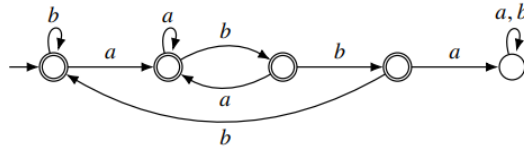


Homework

Question 13: Construct an MSO sentence that defines exactly the languages recognized by the following NFA.



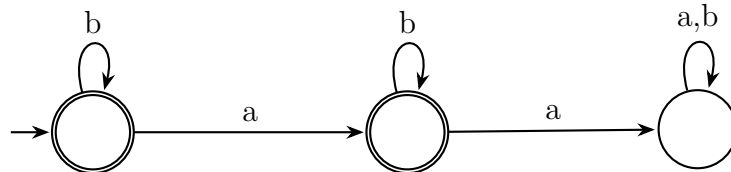
Let L be the language denoted by the regular expression $(a + b)^*abba(a + b)^*$. Then the NFA pictured above recognizes $\bar{L}$; that is, the language of all words that do not contain the subword $abba$. The MSO formula whose models are precisely these words is given below.

Contracted form: $\neg \forall w. a(w) \Rightarrow \forall x. (\text{succ}(w, x) \Rightarrow b(x)) \Rightarrow \forall y. (\text{succ}(x, y) \Rightarrow b(y)) \Rightarrow \forall z. \text{succ}(y, z) \Rightarrow a(z)$

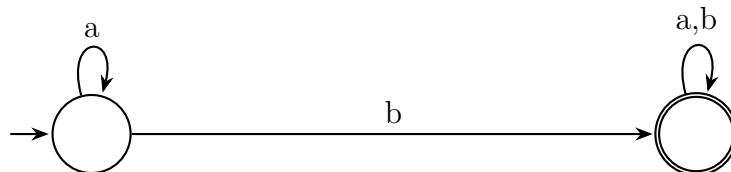
Expanded form: $\neg \forall w [a(w) \Rightarrow \forall x ((\text{succ}(w, x) \Rightarrow b(x)) \Rightarrow \forall y ((\text{succ}(x, y) \Rightarrow b(y)) \Rightarrow \forall z (\text{succ}(y, z) \Rightarrow a(z))))]$

Question 14: Construct NFA that recognize exactly those languages that are defined by the following MSO-sentences.

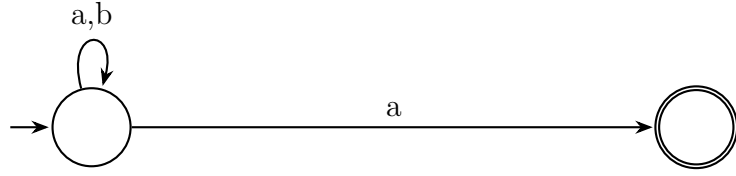
(a) $\forall x. a(x) \Rightarrow \forall y. x < y \Rightarrow b(y)$



(b) $\exists x. b(x) \wedge \forall y. y < x \Rightarrow a(y)$



$$(c) \quad \forall X \exists y \forall x. X(x) \Rightarrow x \leq y \wedge a(y)$$



$$(d) \quad \exists X \exists Y. (\forall z. X(z) \vee Y(z)) \wedge \forall x \forall y. X(x) \wedge Y(y) \Rightarrow x < y \wedge a(x) \wedge b(y)$$

